

BASIC ELECTRICAL ENGINEERING

UNIT I – DC CIRCUITS

This unit introduces the fundamentals of **DC circuits**, covering basic electrical elements, sources, circuit laws, network analysis techniques, and important network theorems. These concepts form the foundation for all electrical and electronics engineering studies.

1. Electrical Circuit Elements

An electrical circuit consists of interconnected elements that control the flow of electric current.

1.1 Resistor (R)

A resistor opposes the flow of electric current and dissipates electrical energy as heat.

Ohm's Law:

$$V = IR$$

where V = voltage, I = current, R = resistance.

Unit: Ohm (Ω)

1.2 Inductor (L)

An inductor stores energy in the form of a magnetic field when current flows through it.

Voltage–Current Relation:

$$V = L (dI/dt)$$

Unit: Henry (H)

1.3 Capacitor (C)

A capacitor stores electrical energy in the form of an electric field.

Current–Voltage Relation:

$$I = C (dV/dt)$$

Unit: Farad (F)

2. Voltage and Current Sources

Electrical sources supply energy to a circuit.

2.1 Independent Sources

Independent sources supply voltage or current independent of other circuit variables.

- Independent voltage source
 - Independent current source
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2.2 Dependent Sources

Dependent sources depend on voltage or current elsewhere in the circuit.

Types:

- Voltage Controlled Voltage Source (VCVS)
 - Voltage Controlled Current Source (VCCS)
 - Current Controlled Voltage Source (CCVS)
 - Current Controlled Current Source (CCCS)
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3. Kirchhoff's Laws

Kirchhoff's laws are fundamental laws used to analyze electrical circuits.

3.1 Kirchhoff's Current Law (KCL)

The algebraic sum of currents at a node is zero.

$$\sum I = 0$$

3.2 Kirchhoff's Voltage Law (KVL)

The algebraic sum of voltages around any closed loop is zero.

$$\sum V = 0$$

4. Analysis of Resistive Circuits

4.1 Mesh Analysis

Mesh analysis uses KVL to find unknown currents in planar circuits.

Steps:

1. Identify meshes
 2. Assign mesh currents
 3. Apply KVL
 4. Solve equations
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4.2 Nodal Analysis

Nodal analysis uses KCL to determine node voltages.

Steps:

1. Select reference node
 2. Assign node voltages
 3. Apply KCL
 4. Solve equations
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4.3 Star–Delta (Y– Δ) Transformations

Star–Delta transformations simplify complex resistor networks.

Used when series–parallel reduction is not possible.

5. Network Theorems

5.1 Superposition Theorem

In a linear circuit, the response due to multiple sources is the sum of responses due to each source acting alone.

5.2 Reciprocity Theorem

In a linear and bilateral network, the current at one branch due to a voltage source in another branch is equal to the current in the second branch when the source is placed in the first branch.

5.3 Thevenin's Theorem

Any linear two-terminal network can be replaced by an equivalent voltage source in series with a resistance.

5.4 Norton's Theorem

Any linear two-terminal network can be replaced by an equivalent current source in parallel with a resistance.

5.5 Maximum Power Transfer Theorem

Maximum power is transferred to the load when load resistance equals source resistance.

$$R_L = R_{Th}$$

6. Numerical Problems

Numerical problems involve:

- Applying KCL and KVL
- Mesh and nodal analysis
- Application of network theorems

Practice of numerical problems strengthens understanding of DC circuit analysis.

UNIT II – AC CIRCUITS

This unit explains the fundamentals of **alternating current (AC) circuits**, sinusoidal quantities, phasor representation, and analysis of single-phase and three-phase AC circuits. These concepts are essential for understanding power systems and electrical machines.

1. AC Fundamentals

Alternating current (AC) is an electric current that periodically changes its magnitude and direction with time.

1.1 Sinusoidal Voltage and Current

A sinusoidal AC voltage or current varies with time according to a sine function.

Expression:

$$V = V_m \sin(\omega t)$$

where V_m = maximum (peak) value, ω = angular frequency, t = time.

1.2 Mathematical and Graphical Representation

- Mathematical form represents AC quantities using equations
 - Graphical form represents variation with time using sine waves
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1.3 Cycle, Time Period and Frequency

- **Cycle:** One complete variation of AC waveform
- **Time Period (T):** Time taken to complete one cycle
- **Frequency (f):** Number of cycles per second

$$f = 1 / T$$

Unit: Hertz (Hz)

1.4 Instantaneous, Peak, Average and RMS Values

- **Instantaneous value:** Value at a particular instant
- **Peak value:** Maximum value of AC quantity
- **Average value:** Mean of values over half cycle
- **RMS value:** Effective value equivalent to DC

For sinusoidal waveform:

- $V_{\text{rms}} = V_m / \sqrt{2}$
 - $I_{\text{rms}} = I_m / \sqrt{2}$
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1.5 Peak Factor and Form Factor

- **Peak factor:** Ratio of peak value to RMS value

$$\text{Peak factor} = V_m / V_{\text{rms}}$$

- **Form factor:** Ratio of RMS value to average value

$$\text{Form factor} = V_{\text{rms}} / V_{\text{avg}}$$

1.6 Phase Difference

Phase difference is the angular difference between two sinusoidal quantities of same frequency.

1.7 Lagging, Leading and In-Phase Quantities

- **Lagging:** Current lags voltage (inductive circuits)
 - **Leading:** Current leads voltage (capacitive circuits)
 - **In phase:** Voltage and current have zero phase difference (resistive circuits)
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2. Phasor Representation

A phasor is a rotating vector representing a sinusoidal quantity.

2.1 Rectangular Representation of Phasors

Phasors are represented using real and imaginary components:

$$A + jB$$

2.2 Polar Representation of Phasors

Phasors are represented by magnitude and angle:

$$|Z| \angle \theta$$

3. Analysis of Single-Phase AC Circuits

3.1 Pure Resistive Circuit (R)

- Voltage and current are in phase
 - Power factor = 1
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3.2 Pure Inductive Circuit (L)

- Current lags voltage by 90°
 - Power factor = 0
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3.3 Pure Capacitive Circuit (C)

- Current leads voltage by 90°
 - Power factor = 0
-

3.4 RL Circuit

- Current lags voltage by angle ϕ
 - Impedance $Z = \sqrt{R^2 + X_L^2}$
-

3.5 RC Circuit

- Current leads voltage by angle ϕ
 - Impedance $Z = \sqrt{R^2 + X_C^2}$
-

3.6 RLC Circuit (Series and Parallel)

- Contains resistance, inductance, and capacitance
 - Can be inductive, capacitive, or resistive
 - Resonance occurs when $X_L = X_C$
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4. Three-Phase Balanced Circuits

Three-phase system consists of three AC voltages of equal magnitude and frequency, displaced by 120° .

4.1 Star (Y) Connection

Relations:

- Line voltage = $\sqrt{3} \times$ Phase voltage
 - Line current = Phase current
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4.2 Delta (Δ) Connection

Relations:

- Line voltage = Phase voltage
- Line current = $\sqrt{3} \times$ Phase current

UNIT III – TRANSFORMERS

This unit deals with **transformers**, which are static electrical devices used to transfer electrical energy from one circuit to another without changing frequency. Transformers play a vital role in power generation, transmission, and distribution systems.

1. Magnetic Circuits

A magnetic circuit is a closed path followed by magnetic flux.

Basic Terms

- **Magnetic flux (Φ):** Total magnetic field produced (Weber)
- **Magnetomotive force (MMF):** Force that produces magnetic flux (NI)
- **Reluctance ($\mathfrak{R}$):** Opposition to magnetic flux

Relation:

$$\Phi = \text{MMF} / \mathfrak{R}$$

Magnetic circuits are analogous to electrical circuits.

2. Transformer

A transformer is a static device that transfers AC power from one circuit to another at the same frequency but different voltage levels through electromagnetic induction.

3. Construction of Transformer

A transformer consists of:

- **Primary winding:** Connected to supply
- **Secondary winding:** Connected to load
- **Magnetic core:** Provides low reluctance path for flux

Types of cores:

- Core type
 - Shell type
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4. Working Principle of Transformer

The working principle of a transformer is **mutual induction**.

- AC supply in primary produces alternating magnetic flux
 - Flux links both windings
 - EMF is induced in secondary winding
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5. Ideal Transformer

An ideal transformer assumes:

- No copper losses
- No core losses
- Infinite permeability
- 100% efficiency

EMF Equation:

$$E = 4.44 f N \Phi_m$$

6. Practical Transformer

In a practical transformer:

- Copper losses occur due to winding resistance
- Core losses occur due to hysteresis and eddy currents
- Leakage flux exists

Hence, efficiency is less than 100%.

7. Equivalent Circuit of Transformer

The equivalent circuit represents transformer losses and impedance using resistances and reactances.

Components:

- Primary resistance and reactance
 - Secondary resistance and reactance
 - Core loss resistance
 - Magnetizing reactance
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8. Losses in Transformers

8.1 Core Losses

- Hysteresis loss
- Eddy current loss

8.2 Copper Losses

- Occur due to resistance of windings
 - Proportional to square of current
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9. Transformer Efficiency

Efficiency is the ratio of output power to input power.

$$\eta = (\text{Output Power} / \text{Input Power}) \times 100$$

Maximum efficiency occurs when:

Copper loss = Core loss

10. Voltage Regulation of Transformer

Voltage regulation is the change in secondary voltage from no-load to full-load.

$$\% \text{ Regulation} = [(V_{\text{no load}} - V_{\text{full load}}) / V_{\text{full load}}] \times 100$$

11. OC and SC Tests on Single-Phase Transformer

11.1 Open Circuit (OC) Test

- Conducted to determine core losses
 - Low voltage applied to HV side
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11.2 Short Circuit (SC) Test

- Conducted to determine copper losses
 - Low voltage applied to LV side
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12. Auto-Transformer

An auto-transformer uses a single winding for both primary and secondary.

Advantages

- Less copper required
- Higher efficiency

Applications

- Voltage control
- Motor starting

13. Three-Phase Transformer Connections

Three-phase transformers are used in power systems.

Common Connections

- Star–Star (Y–Y)
- Delta–Delta (Δ – Δ)
- Star–Delta (Y– Δ)
- Delta–Star (Δ –Y)

UNIT IV – ELECTRICAL MACHINES

This unit deals with **electrical machines**, which convert electrical energy into mechanical energy or vice versa. It covers DC machines, induction motors, and synchronous generators, which are widely used in industrial and power system applications.

1. DC Machines

DC machines are electromechanical devices that operate on direct current. They can work as both **DC generators** and **DC motors**.

2. Construction of DC Machines

The main parts of a DC machine are:

- **Yoke:** Provides mechanical support and magnetic path
- **Field system:** Produces magnetic field
- **Armature core:** Houses armature winding
- **Armature winding:** Conductors placed in armature slots

- **Commutator:** Converts AC to DC (in generator) and DC to AC (in motor)
 - **Brushes:** Collect current from commutator
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3. Armature Windings

Armature winding is the arrangement of conductors in the armature slots.

Types of Armature Windings

- **Lap winding:** Used for high current, low voltage machines
 - **Wave winding:** Used for low current, high voltage machines
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4. DC Generator

A DC generator converts mechanical energy into electrical energy.

Principle of Operation

When a conductor cuts magnetic flux, an EMF is induced in it according to Faraday's laws of electromagnetic induction.

EMF Equation of DC Generator

$$E = (P \Phi Z N) / (60 A)$$

where:

- P = number of poles
- Φ = flux per pole
- Z = total armature conductors
- N = speed in rpm
- A = number of parallel paths

5. DC Motor

A DC motor converts electrical energy into mechanical energy.

Principle of Operation

When a current-carrying conductor is placed in a magnetic field, it experiences a force.

Back EMF

Back EMF is the EMF induced in the armature of a DC motor which opposes the applied voltage.

$$E_b = V - I_a R_a$$

Torque Equation of DC Motor

$$T \propto \Phi I_a$$

Torque is directly proportional to flux and armature current.

6. Rotating Magnetic Field

A rotating magnetic field is produced when three-phase currents flow through stator windings displaced by 120° electrically.

This principle is the basis of operation of AC machines.

7. Induction Motor

An induction motor works on the principle of electromagnetic induction.

7.1 Construction of Induction Motor

Main parts:

- **Stator:** Produces rotating magnetic field
 - **Rotor:** Squirrel cage or slip-ring type
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7.2 Working Principle of Three-Phase Induction Motor

- Three-phase supply produces rotating magnetic field
 - EMF is induced in rotor conductors
 - Rotor current interacts with magnetic field producing torque
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7.3 Single-Phase Induction Motor

Single-phase motors are not self-starting.

Starting methods:

- Split phase
 - Capacitor start
 - Shaded pole motor
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7.4 Torque–Speed Characteristics of Induction Motor

- Torque is zero at synchronous speed
 - Maximum torque occurs at slip less than unity
 - Speed is nearly constant under load
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8. Synchronous Generator

A synchronous generator converts mechanical energy into AC electrical energy.

Construction

- **Stator:** Three-phase armature winding
 - **Rotor:** Field winding excited by DC supply
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Working Principle

- Rotor is excited with DC current
- Rotating magnetic field cuts stator conductors
- AC voltage is induced in stator windings

UNIT V – ELECTRICAL INSTALLATIONS

This unit deals with **electrical installations**, focusing on wiring systems, protection devices, earthing, batteries, and basic electrical calculations. These concepts are essential for safe and efficient use of electrical energy in residential, commercial, and industrial installations.

1. Basic Concept of Wiring Systems

Wiring systems are used to distribute electrical power safely from the supply source to various electrical appliances.

Objectives of Good Wiring

- Safety of users
- Reliability of supply
- Easy maintenance

- Economy and efficiency
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2. Service Mains

Service mains are conductors that connect the consumer installation to the supply authority's distribution system.

Functions:

- Carry electrical power from pole/transformer to consumer premises
 - Provide safe entry of power supply
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3. Meter Board

A meter board houses the energy meter used to measure electrical energy consumed by the user.

Features:

- Mounted at a convenient location
 - Includes main switch and fuses
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4. Distribution Board (DB)

A distribution board distributes electrical power to various circuits in a building.

Functions:

- Protects circuits using MCBs or fuses
 - Ensures proper load distribution
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5. Earthing

Earthing is the process of connecting exposed metallic parts of electrical equipment to the ground.

Purpose of Earthing

- Protects users from electric shock
 - Prevents damage to equipment
 - Maintains system stability
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6. Switch Fuse Unit (SFU)

SFU is a combination of a switch and fuse used for circuit protection and isolation.

Applications:

- Industrial and commercial installations
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7. Miniature Circuit Breaker (MCB)

MCB automatically trips during overload or short circuit conditions.

Advantages:

- Fast operation
 - Reusable
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8. Earth Leakage Circuit Breaker (ELCB)

ELCB disconnects the supply when leakage current flows to earth.

Purpose:

- Protection against electric shock
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9. Moulded Case Circuit Breaker (MCCB)

MCCB is used for higher current ratings and provides overload and short-circuit protection.

Applications:

- Industrial systems
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10. Types of Wires and Cables

Types of Wires

- VIR wires
- PVC insulated wires

Types of Cables

- Single-core cables
 - Multi-core cables
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11. Types of Batteries

Batteries store electrical energy in chemical form.

Types

- Lead-acid batteries
 - Nickel-cadmium batteries
 - Lithium-ion batteries
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12. Important Characteristics of Batteries

- Capacity (Ah)
- Efficiency

- Energy density
 - Charging and discharging rates
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13. Energy Consumption Calculations

Electrical energy consumption is calculated as:

$$\text{Energy (kWh)} = \text{Power (kW)} \times \text{Time (hours)}$$

14. Power Factor Improvement

Power factor improvement reduces power losses and improves efficiency.

Methods:

- Capacitor banks
 - Synchronous condensers
-

15. Battery Backup Calculations

Battery backup time depends on:

- Load power
- Battery voltage
- Battery capacity